

The Drivers of Energy-saving Technical Change

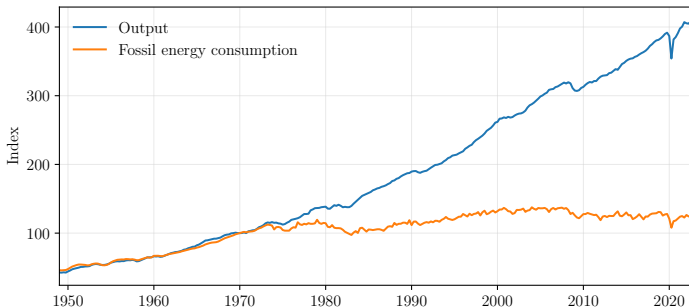
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Energy use and economic growth

- To confront climate challenge, crucial to develop **green technologies**
- Historically, energy use was **key driver** of economic growth



- Since 1970s energy consumption and growth increasingly **decoupled**

This paper

- What can explain the **decoupling** of output and energy use?
- We construct a new measure of **energy-saving technical change**
 - Document a **significant rise** in such technologies . . .
 - . . . coincident with the divergence between output and energy use
- We study the **drivers** underlying energy-saving technical change
 - Based on a semi-structural model of the U.S. economy
 - Account for energy price shocks, TFP and directed technical change
 - Important from a **policy perspective**

Main results

- **Energy price shocks** significantly stimulate energy-saving innovation
 - Directed technical change in response to resource scarcity
 - But also lead to **economic downturn** and crowd-out other innovation
- **TFP shocks** do not lead to energy-saving innovation
 - Output and energy use increase
 - Emissions intensity actually **increases!**
- **Energy-saving technology shocks** key driver of energy-saving innovation
 - Only shock that generates **negative co-movement** between output and energy use
 - Interpreted as sudden breakthroughs in energy efficiency improvements

Main results

- **Energy-saving tech** shocks account for $\sim 60\%$ of variance in energy-saving tech
 - Robust to controlling for structural transformation/service share
 - Captures similar information as shock to green patenting
 - But explain only a small share of changes in energy consumption
 - Such shocks do not create a price signal via higher energy prices
 - **Energy price shocks** key driver of energy consumption
 - Oil shocks, carbon pricing, ...
- ⇒ Need policies that increase **price of emitting** and **support green innovation** to tackle climate change

Conceptual framework

- Starting point is an **aggregate production function**

$$Y_t = F \left(A_t (Z_t K_t)^\alpha (N_t L_t)^{1-\alpha}, A_{et} E_t \right)$$

- Y : business sector output (net value of energy not used in domestic production)
- K, L, E : capital, labor, (fossil) energy
- Z, N : capital utilization, labor effort
- A, A_e : input-saving technology for capital/labor and energy
- α : share parameter

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- We use **nested CES** (following Hassler, Krusell, and Olovsson 2021):

$$Y_t = \left[(1 - \gamma) \left(A_t (Z_t K_t)^\alpha (N_t L_t)^{1-\alpha} \right)^{(\varepsilon-1)/\varepsilon} + \gamma \left(A_{et} E_t \right)^{(\varepsilon-1)/\varepsilon} \right]^{\varepsilon/(\varepsilon-1)}$$

- ε : is the elasticity of substitution between energy and the other inputs

Measurement equations

- Factor demands (assuming competitive factor markets)

$$(1 - \alpha)(1 - \gamma) \left(\frac{A_t (Z_t K_t)^\alpha (N_t L_t)^{1-\alpha}}{Y_t} \right)^{(\varepsilon-1)/\varepsilon} = \frac{W_t N_t L_t}{Y_t} \equiv L_t^{\text{share}}$$
$$\gamma \left(\frac{A_{et} E_t}{Y_t} \right)^{(\varepsilon-1)/\varepsilon} = \frac{P_{et} E_t}{Y_t} \equiv E_t^{\text{share}}$$

- Log-linearizing:

$$\Delta \ln A_t = \Delta \ln Y_t - (\alpha \Delta \ln K_t + (1 - \alpha) \Delta \ln L_t) - \Delta \ln U_t - \frac{\varepsilon}{\varepsilon - 1} \Delta \ln L_t^{\text{share}}$$

$$\Delta \ln A_{et} = \Delta \ln Y_t - \Delta \ln E_t - \frac{\varepsilon}{\varepsilon - 1} \Delta \ln E_t^{\text{share}}$$

- Given α and ε and data for output, inputs and shares can back out the A 's

Estimating the energy complementarity

- The **elasticity of substitution** between energy and other inputs crucial
- Estimate based on the **energy demand schedule**:

$$\Delta \ln Y_t - \Delta \ln E_t = \beta \Delta \ln E_t^{\text{share}} + \mathbf{X}_t' \phi + u_t$$

- Estimate using supply shifters different from energy-saving technical change
- Prime candidates: oil price shocks
- Instrument using Baumeister and Hamilton (2019) and Känzig (2021) oil shocks
- Can back out the elasticity of substitution as $\hat{\varepsilon} = \hat{\beta} / (\hat{\beta} - 1)$

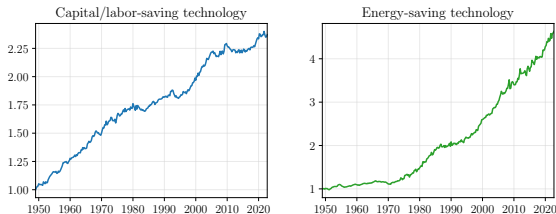
Estimating the energy complementarity

	Baseline	Post-WWII	Post-70s	Pre-GFC
Elasticity of substitution (ε)	0.026 (0.023)	0.030 (0.029)	0.023 (0.022)	0.021 (0.037)
First-stage F statistic	15.63	10.38	18.06	13.17
Observations	181	287	154	229

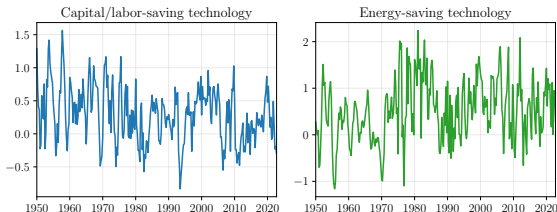
- Energy and other inputs are **highly complementary**
- Robust across different samples, controls included [▶ More](#)
- Estimated elasticity close to structural estimate from Hassler, Krusell, and Olovsson (2021): posterior mean of 0.021

Implied input-saving technical change

(a) Indices



(b) Growth rates



- Strong increase in energy-saving tech in 70s
- Capital/labor-saving tech similar to TFP
- Input-saving technologies very weakly correlated

Identifying drivers of energy-saving technical change

- We want to identify three distinct potential drivers of energy-saving tech
 - Energy price shocks, TFP shocks, energy-saving tech shocks
- Postulate that economy has a VAR representation:

$$\mathbf{A}(L)\mathbf{Y}_t = \mathbf{S}\epsilon_t = \mathbf{u}_t, \quad \text{with } \mathbb{E}[\mathbf{u}_t\mathbf{u}_t'] = \Sigma_u, \text{ and } \mathbf{A}(L) \equiv \mathbf{I} - \mathbf{A}_1L - \dots$$

- We identify the shocks using the max-share approach à la Uhlig (2004)
 - **Idea:** identify shocks that explain maximal FEV in target variable at target horizon

$$\max_{\mathbf{q}_j} \Omega_{i,j}(h) \quad \text{s.t. } \mathbf{q}_j'\mathbf{q}_j = 1,$$

where $\Omega_{i,j}(h) = \frac{\sum_{l=0}^h \mathbf{B}_{i,l} \tilde{\mathbf{S}} \mathbf{q}_j \mathbf{q}_j' \tilde{\mathbf{S}}' \mathbf{B}_{i,l}'}{\sum_{l=0}^h \mathbf{B}_{i,l} \Sigma_u \mathbf{B}_{i,l}'}$ is FEV share of var i to shock j at horizon h

Identifying assumptions

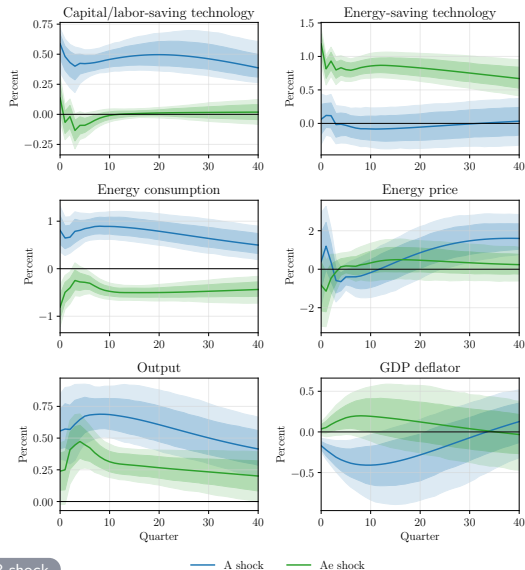
1. Identify **energy price shock** as shock explaining maximum FEV share 4y out
 - This gives us $\mathbf{q}_{pe}$
 - Robust to identifying using oil shocks as external instrument
2. Identify **TFP shock** by maximizing FEV share in capital/labor tech 10y out
 - imposing orthogonality to energy price shock

$$\max_{\mathbf{q}_a} \Omega_{a,a}(40) \quad \text{s.t. } \mathbf{q}'_a \mathbf{q}_a = 1 \text{ and } \mathbf{q}'_a \mathbf{q}_{pe} = 0.$$

3. Identify **energy-saving tech shock** by maximizing FEV in energy tech 10y out
 - imposing orthogonality to energy price shock and capital/labor tech shock

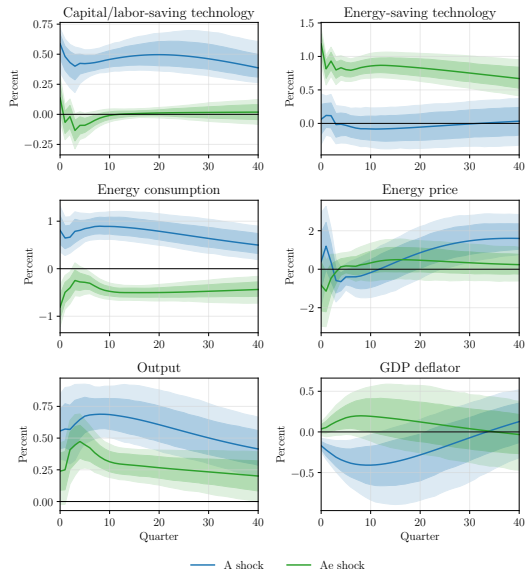
$$\max_{\mathbf{q}_{ae}} \Omega_{ae,ae}(40) \quad \text{s.t. } \mathbf{q}'_{ae} \mathbf{q}_{ae} = 1, \quad \mathbf{q}'_{ae} \mathbf{q}_a = 0 \text{ and } \mathbf{q}'_{ae} \mathbf{q}_{pe} = 0.$$

The impact of input-saving technology shocks



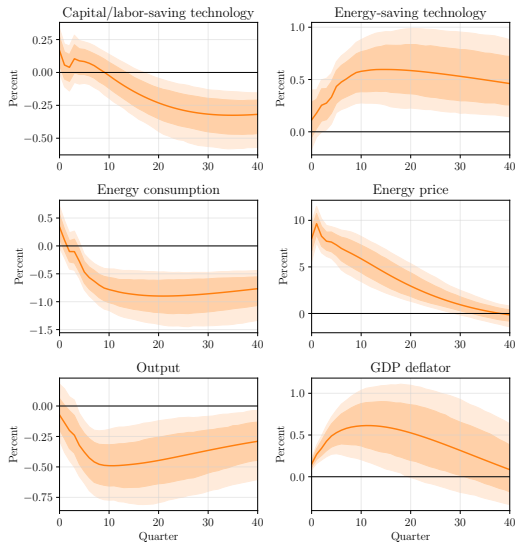
- A shock leads to persistent increase in capital/labor-saving tech
- Energy-saving tech does not change
- Energy consumption goes up significantly
- Output increases, prices fall

The impact of input-saving technology shocks



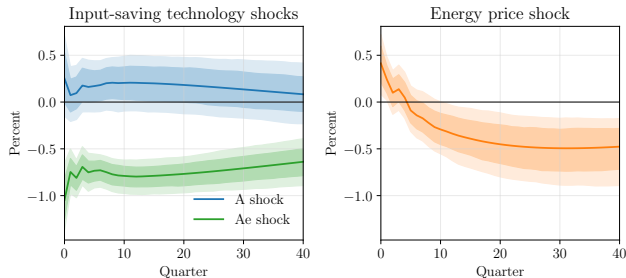
- **Ae** shock leads to persistent increase in energy-saving tech
- Slight crowding out of capital/labor-saving tech
- Energy consumption falls significantly
- Output increases, no big price effect
- Muted energy price response
- Improving energy efficiency does not lead to a fall in energy prices

The impact of energy price shocks



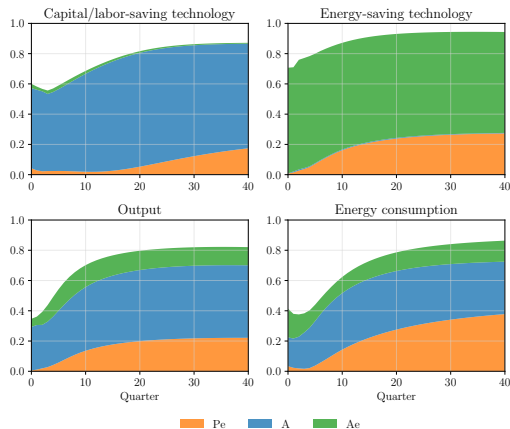
- **Pe** shocks lead to persistent increase in energy prices
- Energy consumption and output fall
- Prices increase
- Leads to substantial energy-saving innovation
- But crowds-out other technologies

Implications for energy intensity



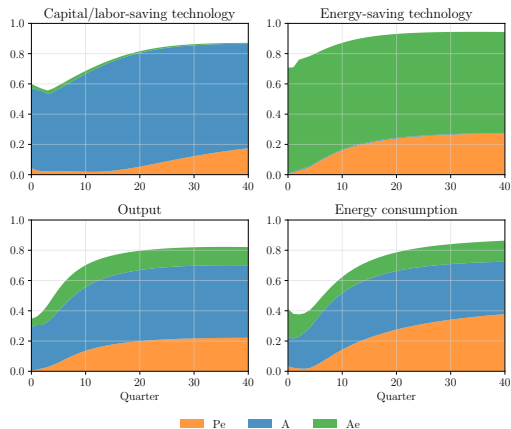
- **Ae** and **Pe** lead to fall in energy intensity
- **A** shock actually increases it!
- But only **Ae** generates negative co-movement between output and energy use

Variance decomposition



- **Ae** shock explain $> 60\%$ in energy-saving tech
- But only explain smaller share of energy consumption (and output)
- **Pe** shocks explain up to 30% in energy-saving tech
- Are also key driver of energy consumption, and non-negligible for output

Variance decomposition

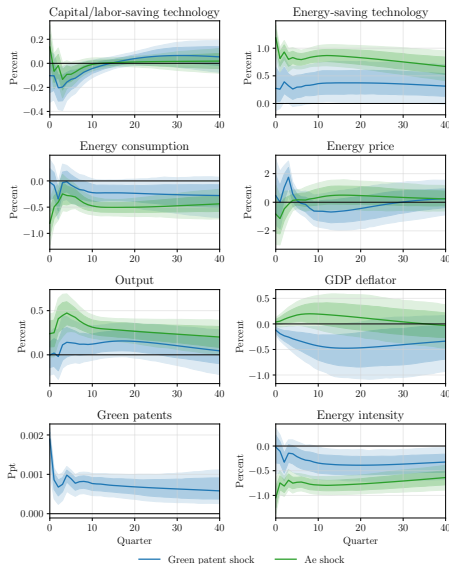


- **A** shock explains bulk of capital/labor-saving tech
- Also explains bulk in output
- But explains little of energy-saving tech

Interpreting energy-saving technology shocks

- **Ae** shocks are key driver of energy-saving technical change
- How should we think about such shocks?
- We interpret the shocks as breakthroughs in the development of energy-saving technologies
- As such, they should be visible in patent data

Interpreting energy-saving technology shocks



- Identify a shock to energy-saving patenting using max-share approach
- Green patenting shock similar to Ae shock
- Generates the same co-movements!
- Energy-saving tech increases, output increases, energy consumption falls $\Rightarrow$ emissions intensity falls

Conclusion

- Study the increasing **divergence** between output and energy use
 - Show that **energy-saving technical change** a crucial factor behind this
 - Energy-saving innovation mainly driven by **energy-saving technology shocks**
 - Only shock that generates negative co-movement between output and energy use
 - **Energy price increases** also lead to energy-saving technical change, but less important
 - However, such increases crucial to explain changes in energy consumption
- ⇒ Need policies that increase **price of emitting** and **support green innovation** to tackle climate change

Thank you!

Related literature

Economic growth and energy use: Kennedy 1964; Acemoglu 2002; Popp 2002; Hanlon 2015; Aghion et al. 2016; Hassler, Krusell, and Olovsson 2021

- Quantify endogenous versus exogenous energy-saving technical change

Technology shocks: Greenwood, Hercowitz, and Krusell 1997; Fisher 2006; Barsky and Sims 2011; Kurmann and Otrok 2013; Kurmann and Sims 2021; Khan et al. 2019; Soofi-Siavash and Moench 2023

- Identify energy-saving tech shock; only shock that yields negative co-movement between output and emissions

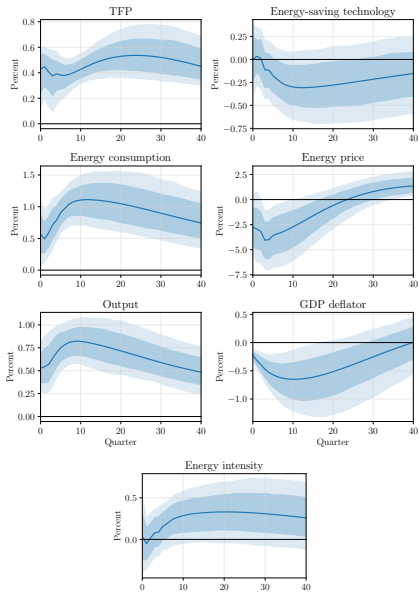
- Output and capital from NIPA
- Labor input based on BLS
- Utilization from Fernald (2014)
- Energy use and prices from EIA
 - Focus on fossil energy
 - Oil, coal and natural gas
 - Quarterly data available since mid-70s, temporally disaggregate annual data before

Estimating the energy complementarity

	Baseline	Känzig supply	BH supply	BH demand
Elasticity of substitution (ε)	0.026 (0.023)	0.012 (0.031)	0.045 (0.028)	0.020 (0.025)
First-stage F statistic	15.63	13.5	10.93	8.20
Observations	181	181	181	181

	No lags	2 lags	4 lags	6 lags
Elasticity of substitution (ε)	0.006 (0.029)	0.016 (0.026)	0.026 (0.023)	0.026 (0.022)
First-stage F statistic	12.41	13.63	15.63	19.21
Observations	185	183	181	179

Do we need to account for input-saving tech shocks?

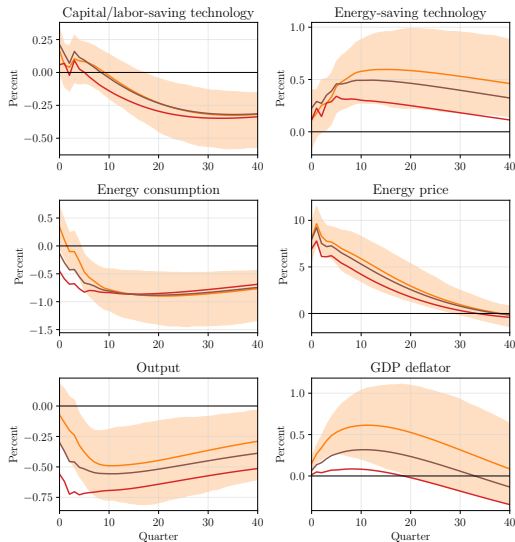


- Identify TFP shock without orthogonalizing
- Looks very similar to our **A** shock
- But importantly looks nothing like a **Ae** shock
- Crucial to disentangle

Identifying energy price shocks

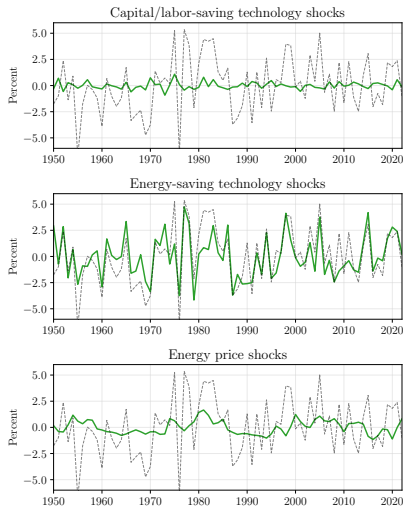
- We identify a shock that explains bulk of variation in energy prices
 - Over post-WWII sample, looks very much like a energy supply shock!
- How robust is this to alternative ways to identify energy price shock?
 - Use Känzig (2021) and Baumeister and Hamilton (2019) oil supply shocks as external instrument

Identifying energy price shocks



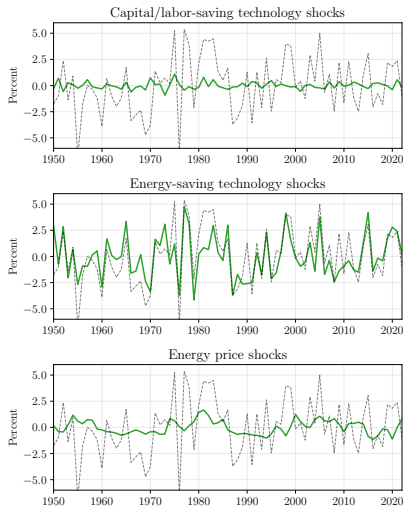
- Results turn out to be robust
- Using Känzig (2021) shock yields very similar results
- Using Baumeister and Hamilton (2019) shock yields weaker price response, but instrument also weaker
- Key challenge: energy price shock instruments only available from 70s

Historical decomposition of energy-saving technology



- Key driver is **Ae** shocks
- Particularly in 90s where many government initiatives for energy efficiency launched
- E.g. Energy Star program, established in 1992

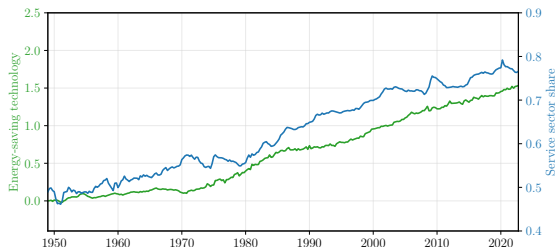
Historical decomposition of energy-saving technology



- **Pe** shocks important in 1970s
- Also in 2014 where energy prices collapsed, mainly for supply related reasons
- Not in early 2000s, which was more driven by energy demand

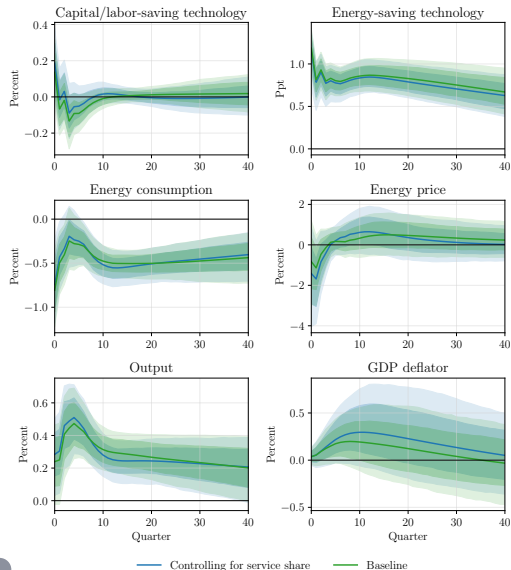
Are we picking up the structural transformation?

- An alternative hypothesis for the decoupling of output and energy consumption is the **structural transformation**
 - Over last decades, service sector has become increasingly important for U.S. economy
 - As a consequence, energy share decreased



- Importantly, service share increased even before 1970s, with no apparent break
- We can also formally control for service sector share ...

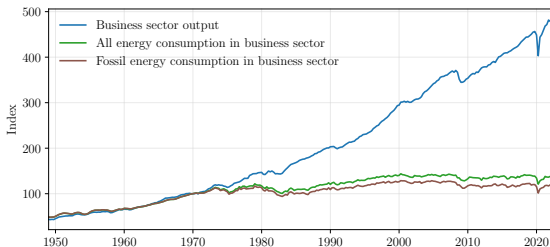
Are we picking up the structural transformation?



- Results are robust to controlling for service share
- Responses essentially unchanged
- Suggests that we are not confounding energy-saving tech with other structural changes to the economy

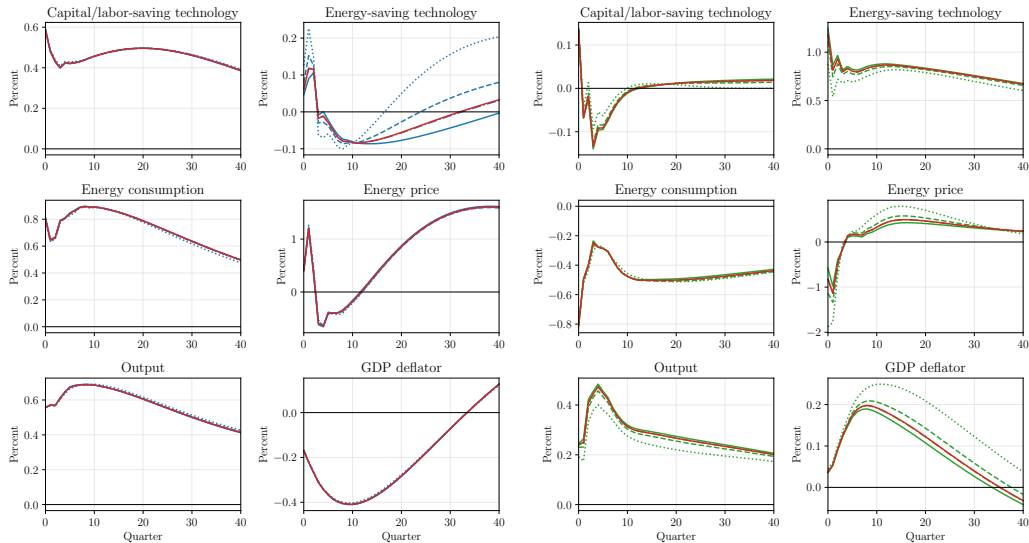
What about renewable energy?

- We have seen a strong divergence between output and fossil fuel use
 - Was there just substitution to renewable energy?

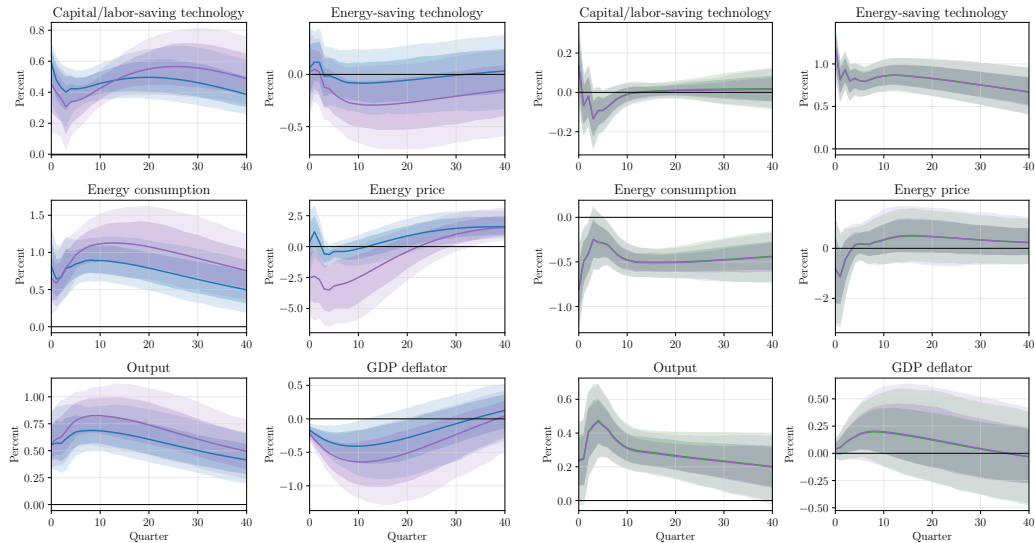


- Renewable energy relatively small!
- Cannot account for the divergence between output and fossil energy use

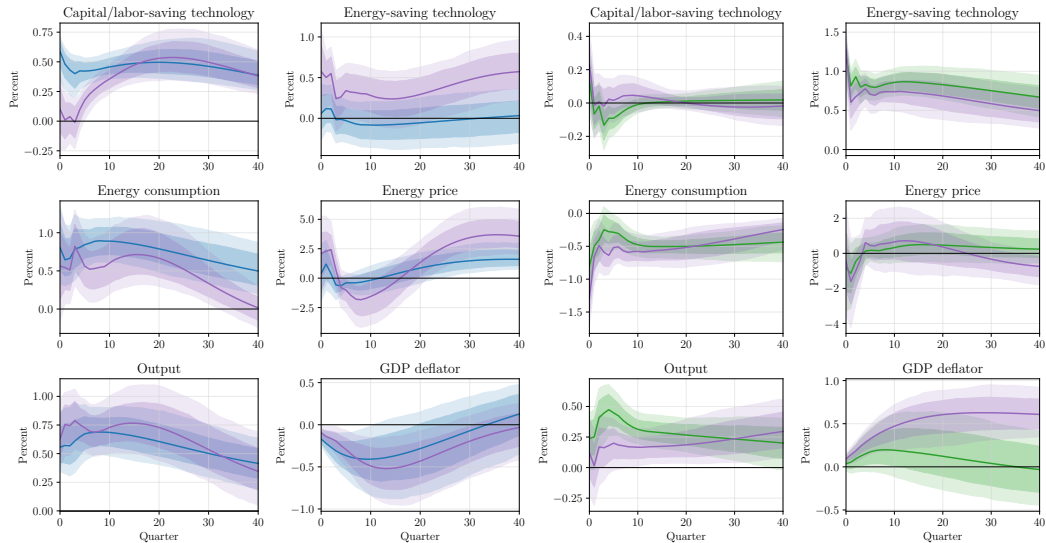
Elasticity



Orthogonalization



Sample period



Sample period

